



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | 100_CAR            |
| Board Name      | custom             |
| Generated with: | STM32CubeMX 6.17.0 |
| Date            | 06/21/2026         |

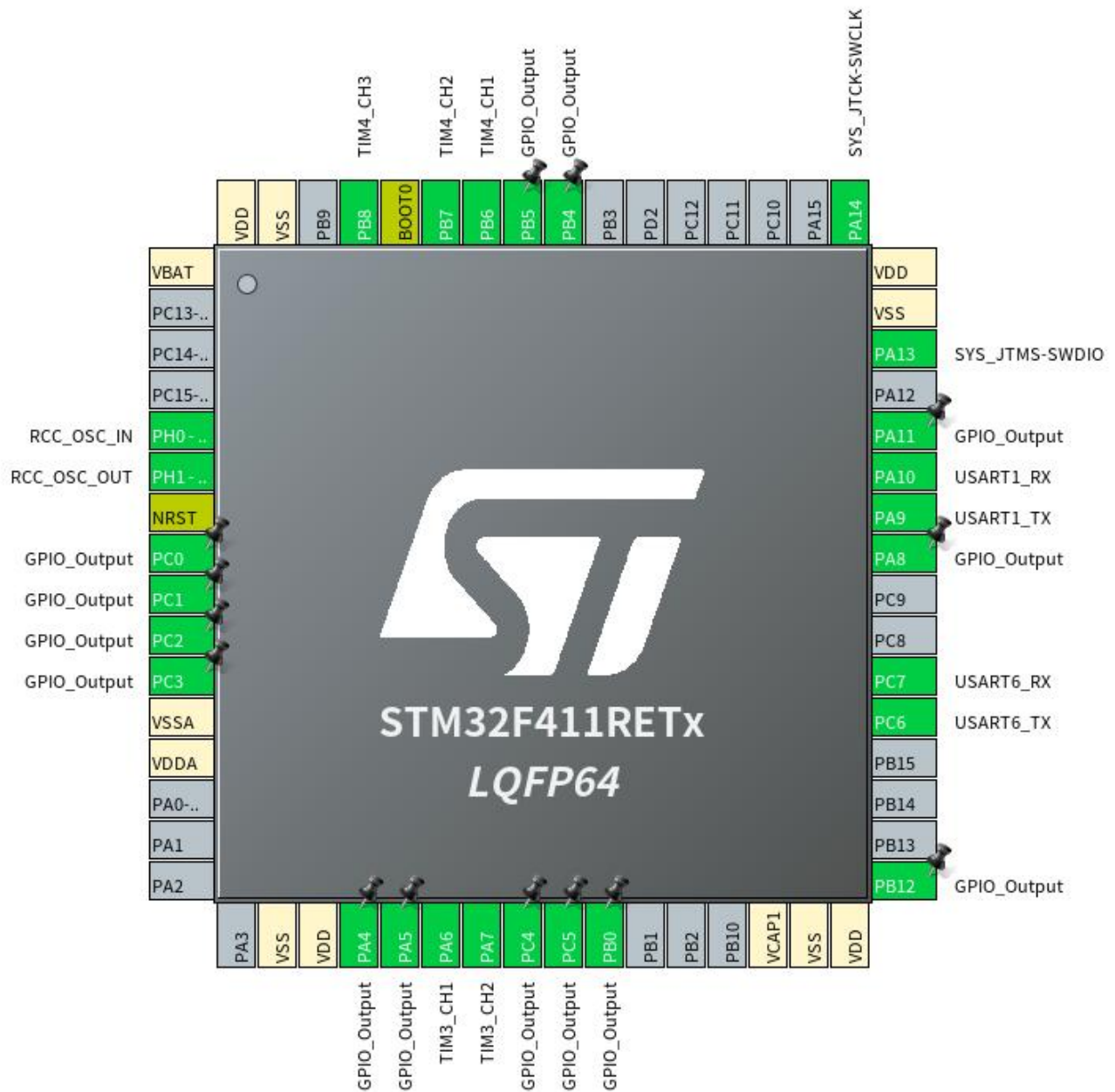
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32F4       |
| MCU Line       | STM32F411     |
| MCU name       | STM32F411RETx |
| MCU Package    | LQFP64        |
| MCU Pin number | 64            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | Arm Cortex-M4 |
|---------|---------------|

## 2. Pinout Configuration



### 3. Pins Configuration

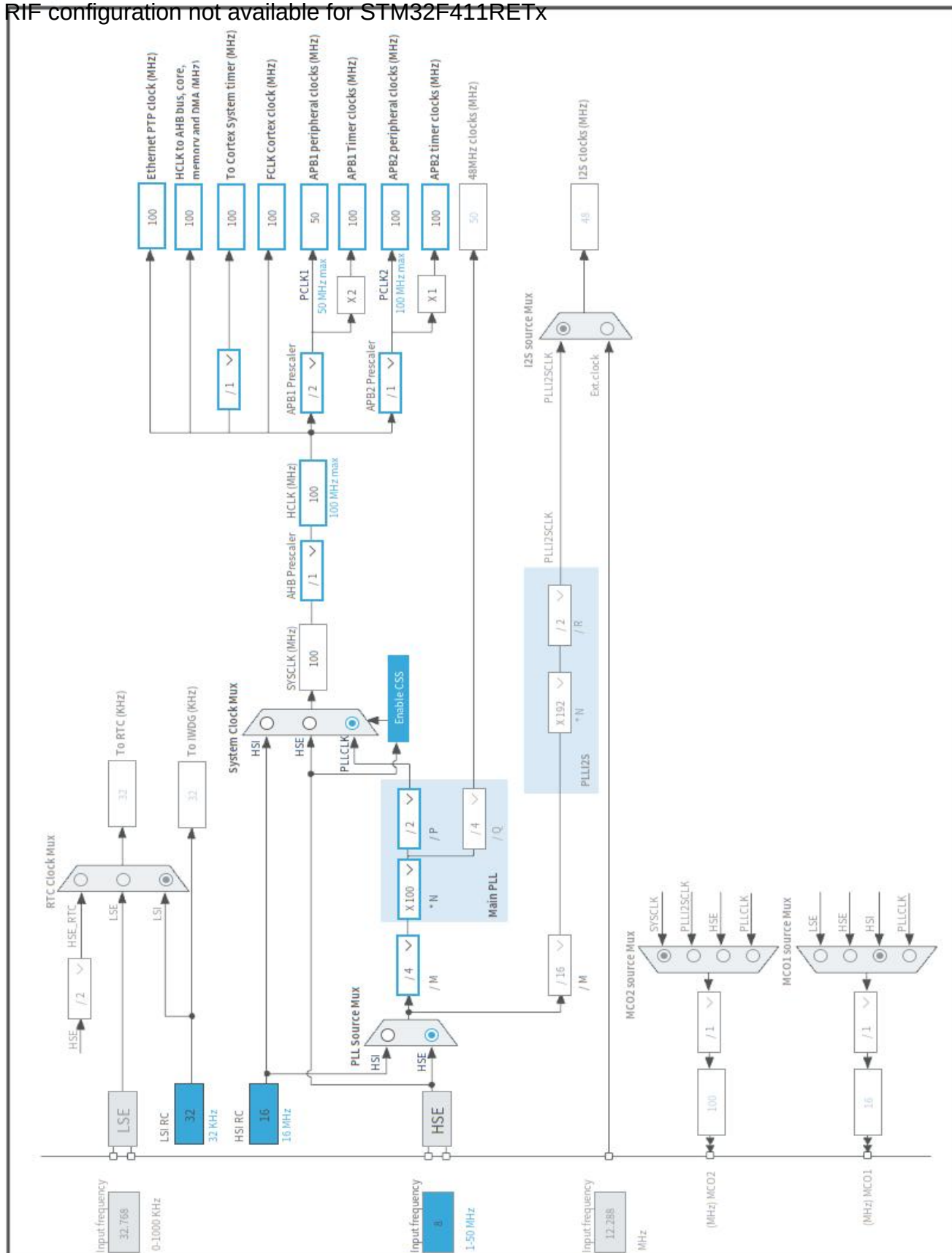
| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label |
|----------------------|---------------------------------------|----------|--------------------------|-------|
| 1                    | VBAT                                  | Power    |                          |       |
| 5                    | PH0 - OSC_IN                          | I/O      | RCC_OSC_IN               |       |
| 6                    | PH1 - OSC_OUT                         | I/O      | RCC_OSC_OUT              |       |
| 7                    | NRST                                  | Reset    |                          |       |
| 8                    | PC0 *                                 | I/O      | GPIO_Output              |       |
| 9                    | PC1 *                                 | I/O      | GPIO_Output              |       |
| 10                   | PC2 *                                 | I/O      | GPIO_Output              |       |
| 11                   | PC3 *                                 | I/O      | GPIO_Output              |       |
| 12                   | VSSA                                  | Power    |                          |       |
| 13                   | VDDA                                  | Power    |                          |       |
| 18                   | VSS                                   | Power    |                          |       |
| 19                   | VDD                                   | Power    |                          |       |
| 20                   | PA4 *                                 | I/O      | GPIO_Output              |       |
| 21                   | PA5 *                                 | I/O      | GPIO_Output              |       |
| 22                   | PA6                                   | I/O      | TIM3_CH1                 |       |
| 23                   | PA7                                   | I/O      | TIM3_CH2                 |       |
| 24                   | PC4 *                                 | I/O      | GPIO_Output              |       |
| 25                   | PC5 *                                 | I/O      | GPIO_Output              |       |
| 26                   | PB0 *                                 | I/O      | GPIO_Output              |       |
| 30                   | VCAP1                                 | Power    |                          |       |
| 31                   | VSS                                   | Power    |                          |       |
| 32                   | VDD                                   | Power    |                          |       |
| 33                   | PB12 *                                | I/O      | GPIO_Output              |       |
| 37                   | PC6                                   | I/O      | USART6_TX                |       |
| 38                   | PC7                                   | I/O      | USART6_RX                |       |
| 41                   | PA8 *                                 | I/O      | GPIO_Output              |       |
| 42                   | PA9                                   | I/O      | USART1_TX                |       |
| 43                   | PA10                                  | I/O      | USART1_RX                |       |
| 44                   | PA11 *                                | I/O      | GPIO_Output              |       |
| 46                   | PA13                                  | I/O      | SYS_JTMS-SWDIO           |       |
| 47                   | VSS                                   | Power    |                          |       |
| 48                   | VDD                                   | Power    |                          |       |
| 49                   | PA14                                  | I/O      | SYS_JTCK-SWCLK           |       |
| 56                   | PB4 *                                 | I/O      | GPIO_Output              |       |
| 57                   | PB5 *                                 | I/O      | GPIO_Output              |       |
| 58                   | PB6                                   | I/O      | TIM4_CH1                 |       |

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label |
|----------------------|---------------------------------------|----------|--------------------------|-------|
| 59                   | PB7                                   | I/O      | TIM4_CH2                 |       |
| 60                   | BOOT0                                 | Boot     |                          |       |
| 61                   | PB8                                   | I/O      | TIM4_CH3                 |       |
| 63                   | VSS                                   | Power    |                          |       |
| 64                   | VDD                                   | Power    |                          |       |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration

RIF configuration not available for STM32F411RETx



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32F4       |
| Line      | STM32F411     |
| MCU       | STM32F411RETx |
| Datasheet | DS10314 Rev6  |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 1.7 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

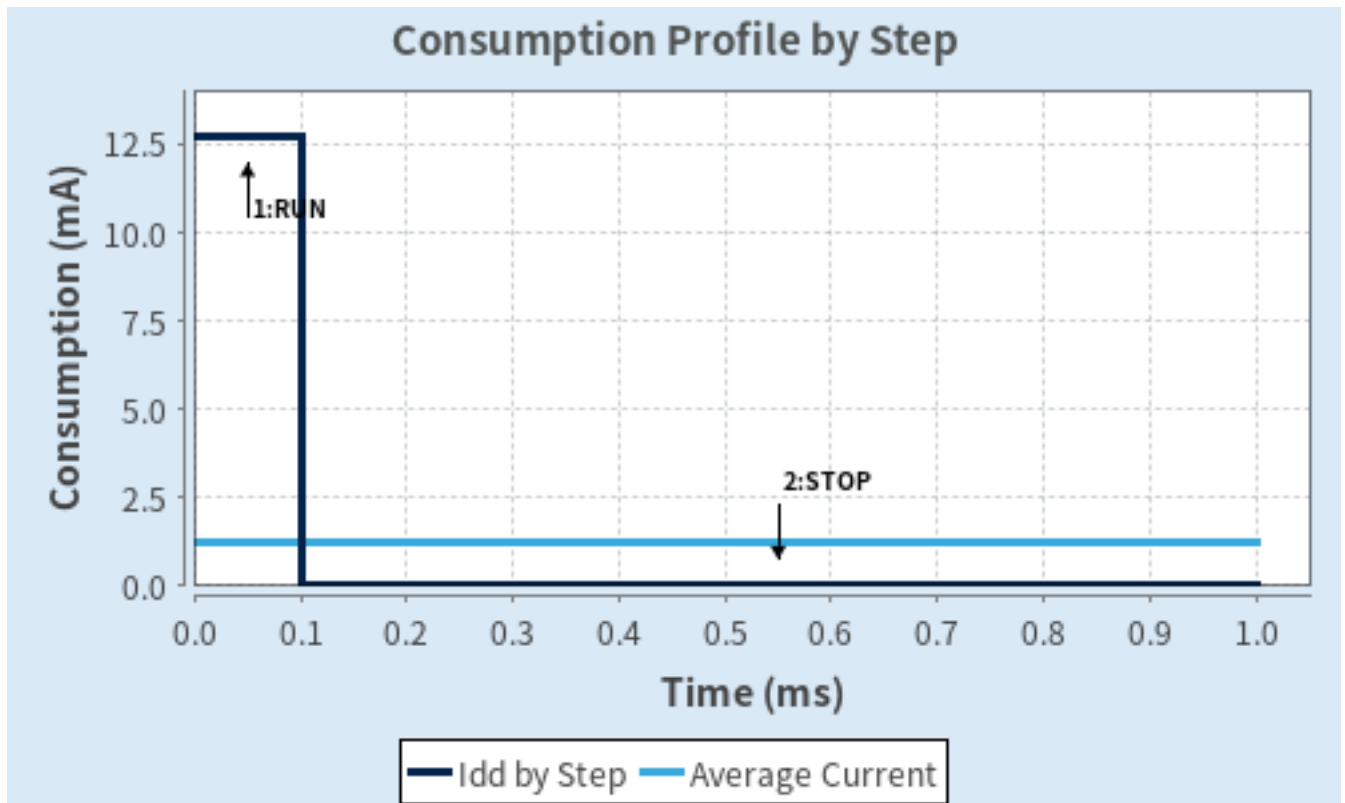
#### 1.4. Sequence

|                               |             |                             |
|-------------------------------|-------------|-----------------------------|
| <b>Step</b>                   | Step1       | Step2                       |
| <b>Mode</b>                   | RUN         | STOP                        |
| <b>Vdd</b>                    | 1.7         | 1.7                         |
| <b>Voltage Source</b>         | Battery     | Battery                     |
| <b>Range</b>                  | Scale1-High | No Scale                    |
| <b>Fetch Type</b>             | SRAM        | n/a                         |
| <b>CPU Frequency</b>          | 100 MHz     | 0 Hz                        |
| <b>Clock Configuration</b>    | HSE PLL     | Regulator_LPLV Flash-PwrDwn |
| <b>Clock Source Frequency</b> | 4 MHz       | 0 Hz                        |
| <b>Peripherals</b>            |             |                             |
| <b>Additional Cons.</b>       | 0 mA        | 0 mA                        |
| <b>Average Current</b>        | 12.7 mA     | 9 $\mu$ A                   |
| <b>Duration</b>               | 0.1 ms      | 0.9 ms                      |
| <b>DMIPS</b>                  | 125.0       | 0.0                         |
| <b>Ta Max</b>                 | 103.99      | 105                         |
| <b>Category</b>               | In DS Table | In DS Table                 |

#### 1.5. Results

|               |                            |                 |             |
|---------------|----------------------------|-----------------|-------------|
| Sequence Time | 1 ms                       | Average Current | 1.28 mA     |
| Battery Life  | 3 months, 19 days, 6 hours | Average DMIPS   | 125.0 DMIPS |

#### 1.6. Chart



## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                         |
|-----------------------------------|-----------------------------------------------|
| Project Name                      | 100_CAR                                       |
| Project Folder                    | /home/minjun/workspace_ondevice_3/ARM/100_CAR |
| Toolchain / IDE                   | STM32CubeIDE                                  |
| Firmware Package Name and Version | STM32Cube FW_F4 V1.28.3                       |
| Application Structure             | Advanced                                      |
| Generate Under Root               | Yes                                           |
| Do not generate the main()        | No                                            |
| Minimum Heap Size                 | 0x200                                         |
| Minimum Stack Size                | 0x400                                         |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | Yes                                   |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name       | Peripheral Instance Name |
|------|---------------------|--------------------------|
| 1    | SystemClock_Config  | RCC                      |
| 2    | MX_GPIO_Init        | GPIO                     |
| 3    | MX_TIM3_Init        | TIM3                     |
| 4    | MX_TIM4_Init        | TIM4                     |
| 5    | MX_TIM11_Init       | TIM11                    |
| 6    | MX_USART6_UART_Init | USART6                   |
| 7    | MX_USART1_UART_Init | USART1                   |

## 3. Peripherals and Middlewares Configuration

### 3.1. RCC

#### High Speed Clock (HSE): BYPASS Clock Source

##### 3.1.1. Parameter Settings:

###### System Parameters:

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Instruction Cache | Enabled            |
| Prefetch Buffer   | Enabled            |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 3 WS (4 CPU cycle) |

###### RCC Parameters:

|                                |          |
|--------------------------------|----------|
| HSI Calibration Value          | 16       |
| TIM Prescaler Selection        | Disabled |
| HSE Startup Timeout Value (ms) | 100      |
| LSE Startup Timeout Value (ms) | 5000     |

###### Power Parameters:

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |
|-------------------------------|---------------------------------|

### 3.2. SYS

#### Debug: Serial Wire

#### Timebase Source: TIM5

### 3.3. TIM3

#### Clock Source : Internal Clock

#### Channel1: PWM Generation CH1

#### Channel2: PWM Generation CH2

##### 3.3.1. Parameter Settings:

###### Counter Settings:

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 100-1 *     |
| Counter Mode                                          | Up          |
| Counter Period (AutoReload Register - 16 bits value ) | 1000-1 *    |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |

###### Trigger Output (TRGO) Parameters:

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
|-----------------------------|--------------------------------------------|

Trigger Event Selection                      Reset (UG bit from TIMx\_EGR)

#### **PWM Generation Channel 1:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

#### **PWM Generation Channel 2:**

|                        |            |
|------------------------|------------|
| Mode                   | PWM mode 1 |
| Pulse (16 bits value)  | 0          |
| Output compare preload | Enable     |
| Fast Mode              | Disable    |
| CH Polarity            | High       |

### **3.4. TIM4**

#### **mode: Clock Source**

**Channel1: Input Capture direct mode**

**Channel2: Input Capture direct mode**

**Channel3: Input Capture direct mode**

#### **3.4.1. Parameter Settings:**

##### **Counter Settings:**

|                                                       |                |
|-------------------------------------------------------|----------------|
| Prescaler (PSC - 16 bits value)                       | <b>100-1 *</b> |
| Counter Mode                                          | Up             |
| Counter Period (AutoReload Register - 16 bits value ) | 65535          |
| Internal Clock Division (CKD)                         | No Division    |
| auto-reload preload                                   | Disable        |

##### **Trigger Output (TRGO) Parameters:**

|                             |                                            |
|-----------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit) | Disable (Trigger input effect not delayed) |
| Trigger Event Selection     | Reset (UG bit from TIMx_EGR)               |

##### **Input Capture Channel 1:**

|                             |             |
|-----------------------------|-------------|
| Polarity Selection          | Rising Edge |
| IC Selection                | Direct      |
| Prescaler Division Ratio    | No division |
| Input Filter (4 bits value) | 0           |

##### **Input Capture Channel 2:**

|                          |             |
|--------------------------|-------------|
| Polarity Selection       | Rising Edge |
| IC Selection             | Direct      |
| Prescaler Division Ratio | No division |

|                                 |             |
|---------------------------------|-------------|
| Input Filter (4 bits value)     | 0           |
| <b>Input Capture Channel 3:</b> |             |
| Polarity Selection              | Rising Edge |
| IC Selection                    | Direct      |
| Prescaler Division Ratio        | No division |
| Input Filter (4 bits value)     | 0           |

### 3.5. TIM11

**mode: Activated**

#### 3.5.1. Parameter Settings:

**Counter Settings:**

|                                                       |                |
|-------------------------------------------------------|----------------|
| Prescaler (PSC - 16 bits value)                       | <b>100-1 *</b> |
| Counter Mode                                          | Up             |
| Counter Period (AutoReload Register - 16 bits value ) | 65535          |
| Internal Clock Division (CKD)                         | No Division    |
| auto-reload preload                                   | Disable        |

### 3.6. USART1

**Mode: Asynchronous**

#### 3.6.1. Parameter Settings:

**Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | <b>9600 *</b>             |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

**Advanced Parameters:**

|                |                      |
|----------------|----------------------|
| Data Direction | Receive and Transmit |
| Over Sampling  | 16 Samples           |

### 3.7. USART6

**Mode: Asynchronous**

#### 3.7.1. Parameter Settings:

**Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

**Advanced Parameters:**

|                |                      |
|----------------|----------------------|
| Data Direction | Receive and Transmit |
| Over Sampling  | 16 Samples           |

### 3.8. FREERTOS

#### Interface: CMSIS\_V2

##### 3.8.1. Config parameters:

**API:**

|              |          |
|--------------|----------|
| FreeRTOS API | CMSIS v2 |
|--------------|----------|

**Versions:**

|                    |        |
|--------------------|--------|
| FreeRTOS version   | 10.3.1 |
| CMSIS-RTOS version | 2.1.3  |

**MPU/FPU:**

|            |          |
|------------|----------|
| ENABLE_MPU | Disabled |
| ENABLE_FPU | Disabled |

**Kernel settings:**

|                               |                 |
|-------------------------------|-----------------|
| USE_PREEMPTION                | Enabled         |
| CPU_CLOCK_HZ                  | SystemCoreClock |
| TICK_RATE_HZ                  | 1000            |
| MAX_PRIORITIES                | 56              |
| USE_SB_COMPLETED_CALLBACK     | 0               |
| USE_MINI_LIST_ITEM            | 1               |
| MINIMAL_STACK_SIZE            | 128             |
| MAX_TASK_NAME_LEN             | 16              |
| USE_16_BIT_TICKS              | Disabled        |
| IDLE_SHOULD_YIELD             | Enabled         |
| USE_MUTEXES                   | Enabled         |
| USE_RECURSIVE_MUTEXES         | Enabled         |
| USE_COUNTING_SEMAPHORES       | Enabled         |
| QUEUE_REGISTRY_SIZE           | 8               |
| USE_APPLICATION_TASK_TAG      | Disabled        |
| ENABLE_BACKWARD_COMPATIBILITY | Enabled         |

|                                   |          |
|-----------------------------------|----------|
| USE_PORT_OPTIMISED_TASK_SELECTION | Disabled |
| USE_TICKLESS_IDLE                 | Disabled |
| USE_TASK_NOTIFICATIONS            | Enabled  |
| RECORD_STACK_HIGH_ADDRESS         | Disabled |

#### Memory management settings:

|                          |                  |
|--------------------------|------------------|
| Memory Allocation        | Dynamic / Static |
| TOTAL_HEAP_SIZE          | 15360            |
| Memory Management scheme | heap_4           |

#### Hook function related definitions:

|                              |          |
|------------------------------|----------|
| USE_IDLE_HOOK                | Disabled |
| USE_TICK_HOOK                | Disabled |
| USE_MALLOC_FAILED_HOOK       | Disabled |
| USE_DAEMON_TASK_STARTUP_HOOK | Disabled |
| CHECK_FOR_STACK_OVERFLOW     | Disabled |

#### Run time and task stats gathering related definitions:

|                                |          |
|--------------------------------|----------|
| GENERATE_RUN_TIME_STATS        | Disabled |
| USE_TRACE_FACILITY             | Enabled  |
| USE_STATS_FORMATTING_FUNCTIONS | Disabled |

#### Co-routine related definitions:

|                           |          |
|---------------------------|----------|
| USE_CO_ROUTINES           | Disabled |
| MAX_CO_ROUTINE_PRIORITIES | 2        |

#### Software timer definitions:

|                        |         |
|------------------------|---------|
| USE_TIMERS             | Enabled |
| TIMER_TASK_PRIORITY    | 2       |
| TIMER_QUEUE_LENGTH     | 10      |
| TIMER_TASK_STACK_DEPTH | 256     |

#### Interrupt nesting behaviour configuration:

|                                        |    |
|----------------------------------------|----|
| LIBRARY_LOWEST_INTERRUPT_PRIORITY      | 15 |
| LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY | 5  |

#### Added with 10.2.1 support:

|                            |          |
|----------------------------|----------|
| MESSAGE_BUFFER_LENGTH_TYPE | size_t   |
| USE_POSIX_ERRNO            | Disabled |

#### CMSIS-RTOS V2 flags:

|                               |         |
|-------------------------------|---------|
| USE_OS2_THREAD_SUSPEND_RESUME | Enabled |
| USE_OS2_THREAD_ENUMERATE      | Enabled |
| USE_OS2_EVENTFLAGS_FROM_ISR   | Enabled |
| USE_OS2_THREAD_FLAGS          | Enabled |
| USE_OS2_TIMER                 | Enabled |
| USE_OS2_MUTEX                 | Enabled |

### 3.8.2. Include parameters:

#### **Include definitions:**

|                              |          |
|------------------------------|----------|
| vTaskPrioritySet             | Enabled  |
| uxTaskPriorityGet            | Enabled  |
| vTaskDelete                  | Enabled  |
| vTaskCleanUpResources        | Disabled |
| vTaskSuspend                 | Enabled  |
| vTaskDelayUntil              | Enabled  |
| vTaskDelay                   | Enabled  |
| xTaskGetSchedulerState       | Enabled  |
| xTaskResumeFromISR           | Enabled  |
| xQueueGetMutexHolder         | Enabled  |
| xSemaphoreGetMutexHolder     | Disabled |
| pcTaskGetTaskName            | Disabled |
| uxTaskGetStackHighWaterMark  | Enabled  |
| xTaskGetCurrentTaskHandle    | Enabled  |
| eTaskGetState                | Enabled  |
| xEventGroupSetBitFromISR     | Disabled |
| xTimerPendFunctionCall       | Enabled  |
| xTaskAbortDelay              | Disabled |
| xTaskGetHandle               | Disabled |
| uxTaskGetStackHighWaterMark2 | Disabled |

### 3.8.3. Advanced settings:

#### **Newlib settings (see parameter description first):**

USE\_NEWLIB\_REENTRANT                      **Enabled \***

#### **Project settings (see parameter description first):**

Use FW pack heap file                      Enabled

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP     | Pin           | Signal         | GPIO mode                    | GPIO pull/up pull down      | Max Speed   | User Label |
|--------|---------------|----------------|------------------------------|-----------------------------|-------------|------------|
| RCC    | PH0 - OSC_IN  | RCC_OSC_IN     | n/a                          | n/a                         | n/a         |            |
|        | PH1 - OSC_OUT | RCC_OSC_OUT    | n/a                          | n/a                         | n/a         |            |
| SYS    | PA13          | SYS_JTMS-SWDIO | n/a                          | n/a                         | n/a         |            |
|        | PA14          | SYS_JTCK-SWCLK | n/a                          | n/a                         | n/a         |            |
| TIM3   | PA6           | TIM3_CH1       | Alternate Function Push Pull | No pull-up and no pull-down | Low         |            |
|        | PA7           | TIM3_CH2       | Alternate Function Push Pull | No pull-up and no pull-down | Low         |            |
| TIM4   | PB6           | TIM4_CH1       | Alternate Function Push Pull | No pull-up and no pull-down | Low         |            |
|        | PB7           | TIM4_CH2       | Alternate Function Push Pull | No pull-up and no pull-down | Low         |            |
|        | PB8           | TIM4_CH3       | Alternate Function Push Pull | No pull-up and no pull-down | Low         |            |
| USART1 | PA9           | USART1_TX      | Alternate Function Push Pull | No pull-up and no pull-down | Very High * |            |
|        | PA10          | USART1_RX      | Alternate Function Push Pull | No pull-up and no pull-down | Very High * |            |
| USART6 | PC6           | USART6_TX      | Alternate Function Push Pull | No pull-up and no pull-down | Very High * |            |
|        | PC7           | USART6_RX      | Alternate Function Push Pull | No pull-up and no pull-down | Very High * |            |
| GPIO   | PC0           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PC1           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PC2           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PC3           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PA4           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PA5           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PC4           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PC5           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PB0           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PB12          | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PA8           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PA11          | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PB4           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |
|        | PB5           | GPIO_Output    | Output Push Pull             | No pull-up and no pull-down | Low         |            |

#### **4.2. DMA configuration**

nothing configured in DMA service

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                                                    | Enable | Preenmption Priority | SubPriority |
|--------------------------------------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                                             | true   | 0                    | 0           |
| Hard fault interrupt                                               | true   | 0                    | 0           |
| Memory management fault                                            | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault                               | true   | 0                    | 0           |
| Undefined instruction or illegal state                             | true   | 0                    | 0           |
| System service call via SWI instruction                            | true   | 0                    | 0           |
| Debug monitor                                                      | true   | 0                    | 0           |
| Pendable request for system service                                | true   | 15                   | 0           |
| System tick timer                                                  | true   | 15                   | 0           |
| TIM4 global interrupt                                              | true   | 5                    | 0           |
| USART1 global interrupt                                            | true   | 5                    | 0           |
| TIM5 global interrupt                                              | true   | 15                   | 0           |
| USART6 global interrupt                                            | true   | 5                    | 0           |
| PVD interrupt through EXTI line 16                                 | unused |                      |             |
| Flash global interrupt                                             | unused |                      |             |
| RCC global interrupt                                               | unused |                      |             |
| TIM1 trigger and commutation interrupts and TIM11 global interrupt | unused |                      |             |
| TIM3 global interrupt                                              | unused |                      |             |
| FPU global interrupt                                               | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Pre-fetch fault, memory access fault    | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | false                | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | false                | false            |
| System tick timer                       | false                             | false                | true             |
| TIM4 global interrupt                   | false                             | true                 | true             |
| USART1 global interrupt                 | false                             | true                 | true             |
| TIM5 global interrupt                   | false                             | true                 | true             |
| USART6 global interrupt                 | false                             | true                 | true             |

**\* User modified value**

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

#### Middleware

FREERTOS 

#### System Core

DMA

GPIO 

NVIC 

RCC 

SYS 

#### Analog

#### Timers

TIM3 

TIM4 

TIM11 

#### Connectivity

USART1 

USART6 

#### Multimedia

#### Computing

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32f411_bsdI.zip">https://www.st.com/resource/en/bsdl_model/stm32f411_bsdI.zip</a>                                                                                                                                                                                                 |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32f411_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32f411_ibis.zip</a>                                                                                                                                                                                                 |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32f4-svd.zip">https://www.st.com/resource/en/svd/stm32f4-svd.zip</a>                                                                                                                                                                                                                     |
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| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                                     |
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| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                                                             |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf">https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf</a>                                                                                                                                     |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf">https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf</a>                                                                                                                                   |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32f4x1.pdf">https://www.st.com/resource/en/flyer/flstm32f4x1.pdf</a>                                                                                                                                                                                                                 |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                                             |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32trust.pdf">https://www.st.com/resource/en/flyer/flstm32trust.pdf</a>                                                                                                                                                                                                               |
| Product Certifications  | <a href="https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf">https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf</a>                                                                                                                                                     |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf</a> |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf</a>                                                               |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an2945-stm8s-and-">https://www.st.com/resource/en/application_note/an2945-stm8s-and-</a>                                                                                                                                                                                       |

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